The One-Phase Stefan Problem: A Closed Neumann Root, Lambert-W Endpoints, and an Exact Energy Ledger

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Abstract

We give a source-local theorem for the dimensionless one-phase Stefan problem. The Neumann similarity ansatz reduces the moving boundary to the strictly increasing equation $F(\lambda) = \sqrt{\pi}\lambda e^{\lambda^2} \operatorname{erf} \lambda = \mathrm{Ste}$. This proves a unique positive root, supplies a five-term small-Ste inverse series and a two-sided large-Ste Lambert-W enclosure, and closes the wall/interface flux partition. Integrating the heat equation over the moving interval gives an exact sensible-plus-latent energy ledger. Zero superheat, zero diffusivity, and zero latent heat are labelled singular rescalings. The result is hydrodynamic/free-boundary analysis, not an arithmetic determinant or a Hilbert–Pólya construction.

1 Frozen model and similarity reduction

We scale the wall superheat, melting temperature, and thermal diffusivity to one. The liquid occupies $0 < x < s(t)$ and satisfies

$$u_t = u_{xx}, \qquad u(0,t) = 1, \qquad u(s(t),t) = 0, \qquad \beta s'(t) = -u_x(s(t)^-,t), \qquad s(0) = 0, \tag{1}$$

where $\beta = \mathrm{Ste}^{-1} > 0$ is the inverse Stefan number. With $\eta = x/(2\sqrt{t})$, the Neumann form is

$$s(t) = 2\lambda\sqrt{t}, \qquad u(x,t) = 1 - \frac{\operatorname{erf}(\eta)}{\operatorname{erf}(\lambda)}. \tag{2}$$

The interface derivative is $-u_x(s(t)^-,t) = e^{-\lambda^2}/(\sqrt{\pi t}\operatorname{erf}\lambda)$. Consequently the Stefan condition is equivalent to

$$F(\lambda) := \sqrt{\pi}\lambda e^{\lambda^2} \operatorname{erf}\lambda = \mathrm{Ste}. \tag{3}$$

Theorem 1 (unique Neumann root and endpoint atlas)**.** *For every* Ste > 0*, equation* (3) *has exactly one* $\lambda > 0$*. Indeed,* $F(0) = 0$*,* $F(\lambda) \to \infty$*, and*

$$F'(\lambda) = \sqrt{\pi}e^{\lambda^2} \operatorname{erf}\lambda(1 + 2\lambda^2) + 2\lambda > 0. \tag{4}$$

The corresponding profile (2) *solves* (1) *for* $t > 0$*. Writing* $z = \mathrm{Ste}$*, its small-*z *inverse is*

$$\lambda^2 = \frac{z}{2} - \frac{z^2}{6} + \frac{7z^3}{90} - \frac{79z^4}{1890} + \frac{689z^5}{28350} + O(z^6), \tag{5}$$

and, for $z > 1$*,*

$$\frac{1}{2}W\left(\frac{2z^2}{\pi}\right) < \lambda^2 < \frac{1}{2}W\left(\frac{2(z+1)^2}{\pi}\right). \tag{6}$$

Thus $\lambda^2 \sim \frac{1}{2}W(2z^2/\pi)$ *as* $z \to \infty$*.*

Proof. Substitution of (2) into the heat equation uses $\frac{\mathrm{d}}{\mathrm{d}\eta}\operatorname{erf}\eta = 2e^{-\eta^2}/\sqrt{\pi}$; the two Dirichlet values are immediate. Differentiating F gives (4), while its endpoint limits follow from the standard limits of erf. Expanding

$$F(\sqrt{x}) = 2x + \frac{4}{3}x^2 + \frac{8}{15}x^3 + \frac{16}{105}x^4 + \frac{32}{945}x^5 + O(x^6)$$

and formal reversion gives (5). Finally, for $\lambda > 0$, $0 < \operatorname{erfc}\lambda < e^{-\lambda^2}/(\sqrt{\pi}\lambda)$, so

$$z < \sqrt{\pi\lambda^2}e^{\lambda^2} < z + 1.$$

The function $x \mapsto \sqrt{\pi x}e^x$ is increasing, and inversion with the principal Lambert branch proves (6). ∎

2 Flux partition and the moving-domain energy identity

The wall and interface fluxes have a common $t^{-1/2}$ singularity:

$$J_{\text{wall}}(t) = -u_x(0,t) = \frac{1}{\sqrt{\pi t}\,\operatorname{erf}\lambda}, \qquad J_{\text{int}}(t) = -u_x(s(t)^-,t) = e^{-\lambda^2} J_{\text{wall}}(t). \tag{7}$$

Thus $J_{\text{int}}/J_{\text{wall}} = e^{-\lambda^2}$ exactly. Integrating $u_t = u_{xx}$ on the moving interval (the boundary value $u(s(t),t) = 0$ removes the Reynolds endpoint term) gives

$$\int_0^t J_{\text{wall}}(\tau)\,\mathrm{d}\tau = \int_0^{s(t)} u(x,t)\,\mathrm{d}x + \beta s(t). \tag{8}$$

The elementary antiderivative $\int_0^\lambda \operatorname{erf}\eta\,\mathrm{d}\eta = \lambda\operatorname{erf}\lambda + (e^{-\lambda^2} - 1)/\sqrt{\pi}$ therefore yields

$$S(t) := \int_0^{s(t)} u\,\mathrm{d}x = \frac{2\sqrt{t}(1 - e^{-\lambda^2})}{\sqrt{\pi}\operatorname{erf}\lambda}, \tag{9}$$

$$L(t) := \beta s(t) = 2\beta\lambda\sqrt{t} = \frac{2e^{-\lambda^2}\sqrt{t}}{\sqrt{\pi}\operatorname{erf}\lambda}, \tag{10}$$

$$I(t) := \int_0^t J_{\text{wall}}(\tau)\,\mathrm{d}\tau = \frac{2\sqrt{t}}{\sqrt{\pi}\operatorname{erf}\lambda} = S(t) + L(t). \tag{11}$$

This is an exact sensible-plus-latent energy ledger, not a fitted numerical balance. The eight rational probes in the accompanying receipt cover both terms and the Lambert enclosure.

3 Singular limits, source boundary, and audit

As $\beta \to \infty$ (small Ste), $\lambda \to 0$ by (5); the normalized interface speed vanishes. As $\beta \to 0^+$ (large Ste), $\lambda \to \infty$ with (6). Zero superheat makes the chosen normalization undefined, and zero dimensional diffusivity $\kappa = 0$ collapses the similarity length $2\lambda\sqrt{\kappa t}$ (here κ denotes the dimensional diffusivity before the displayed scaling). The zero-latent limit $L = 0$ is $\mathrm{Ste} = \infty$ and requires a separate rescaling, as emphasized in the small-latent-heat analysis of Addison–Howison–King [1]; no finite-λ Stefan interface is claimed. These boundaries are distinct from the fixed-domain Barenblatt and KPP families (C207 and C202): here the defining advance is a moving boundary and the exact ledger (11).

`(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).`

Hence `overall=ROUTE_A_REJECTED` and `route_b_invocation_allowed=false`. The source heat clock is not target continuation/divisor/counting law and is not an A3 analytic-structure match. The scope lock is

`NO_BAD_EULER_OR_ROOT_NUMBER.`

No target prime or zero table, Euler factor, root number, automorphy statement, functional equation, or Hilbert–Pólya operator is introduced.

Source note

The classical free-boundary setting and small-latent-heat asymptotics are documented by Addison, Howison, and King [1]; phase-change and Neumann conventions are treated in Gupta [2]; two-phase stability background is given by Rubinstein [3]. These references are source metadata, not a priority or arithmetic claim.

References

[1] J. A. Addison, S. D. Howison, and J. R. King, Ray methods for Free Boundary Problems, Oxford author manuscript (2005), smallstefan.pdf.

[2] S. C. Gupta, *The Classical Stefan Problem: Basic Concepts, Modelling and Analysis*, Elsevier, North-Holland Series in Applied Mathematics and Mechanics, Volume 45 (2003), ISBN 978-0-444-51086-0.

[3] L. I. Rubinstein, Global Stability of the Neumann Solution of the Two-phase Stefan Problem, *IMA Journal of Applied Mathematics* 28(3), 287–299 (1982). DOI: 10.1093/imamat/28.3.287.

Declarations

Scope. `NO_BAD_EULER_OR_ROOT_NUMBER`; no target arithmetic or operator claim. **Data and code.** The theorem, ledger, independent checks, and build instructions are released with HCS-C226. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked displayed claims and metadata. This is not external peer review.